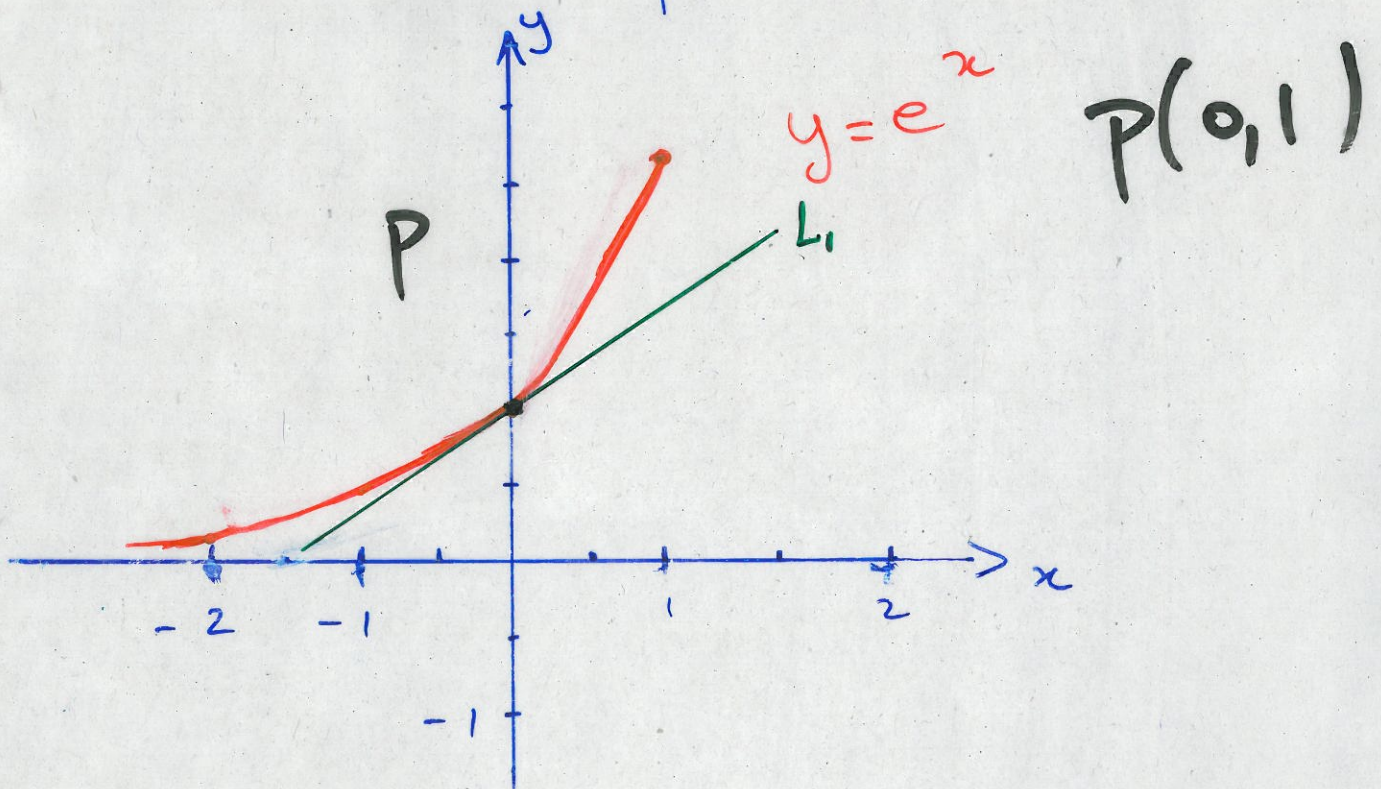


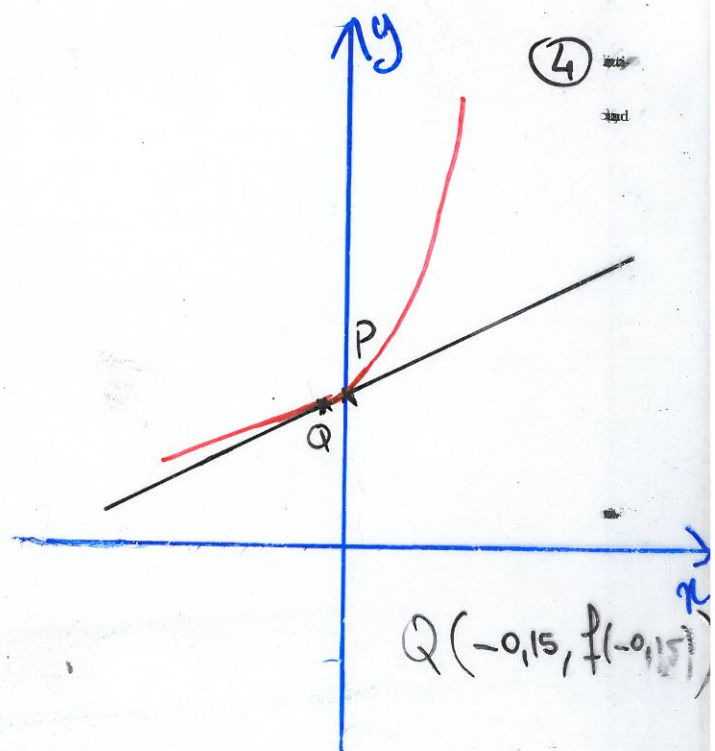
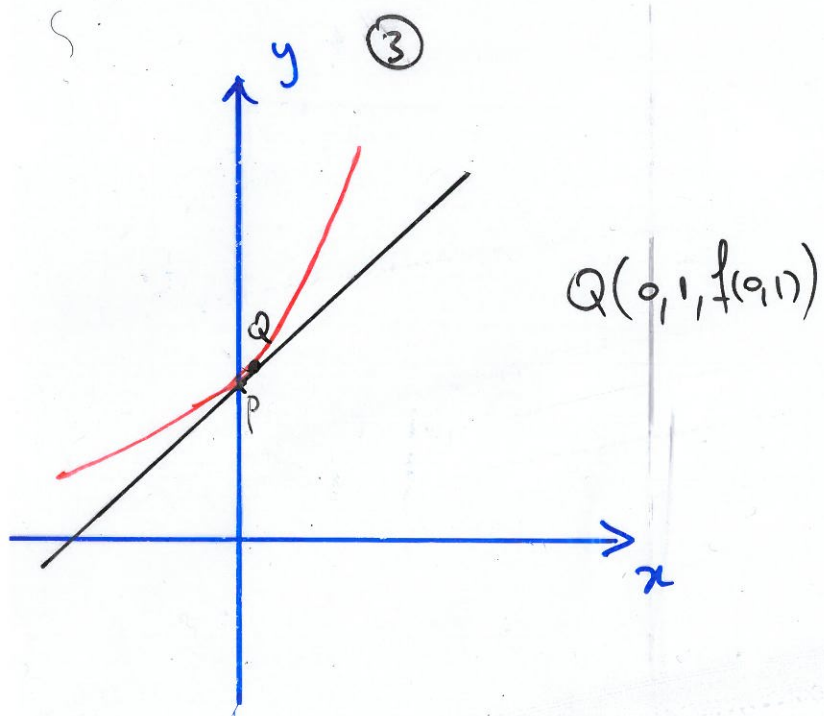
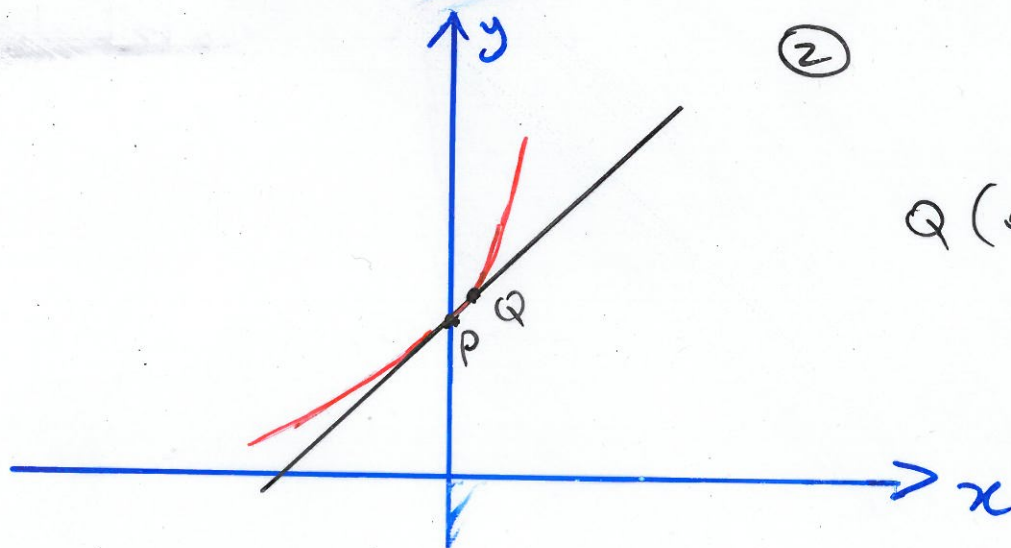
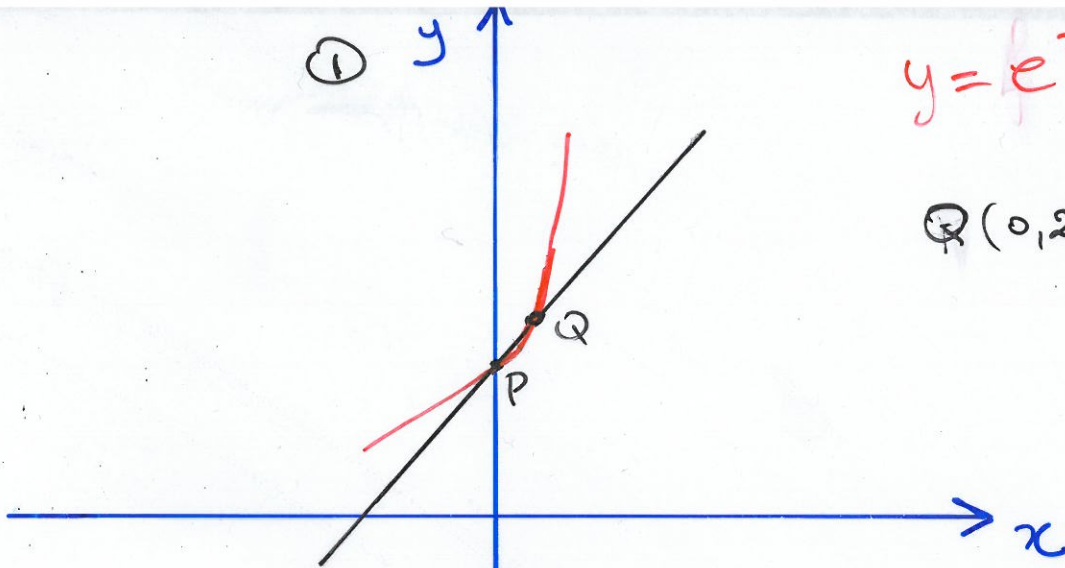
2.1 The tangent and velocity problems

Given a curve $y = f(x)$, we would like to estimate the slope of the tangent to $y = f(x)$ at (x_0, y_0) .

For instance, let $f(x) = e^x$



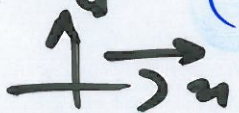
let estimate the slope of L_1 by using the slope of the secant line PQ , with $P(0, 1)$ and $Q(x, y)$ on $y = f(x)$



We obtain the following table :

$$m_{PQ} = \frac{f(x) - f(x_0)}{x - x_0} = \frac{f(x) - 1}{x} = \frac{e^x - 1}{x}$$

$$\Leftrightarrow m_{PQ} = \frac{f(x_0+h) - f(x_0)}{(x_0+h) - x_0} = \frac{f(h) - f(0)}{h} \quad \text{with } x_0 = 0$$




$h > 0$	m_{PQ}
1	1,7182
0,5	1,2974
0,4	1,2295
0,3	1,1661
0,25	1,1361
0,17	1,0900
0,1	1,0517
0,08	1,0410
0,001	1,0005
0,0001	1,00005

$h < 0$	m_{PQ}
-10^{-7}	1,000
-10^{-5}	1,00000
-0,0001	0,9999
-0,001	0,9995
-0,01	0,9950
-0,1	0,9516
-0,15	0,9286
-0,27	0,8763
-0,4	0,8241
-0,6	0,7519
-1	0,6321

From the table as Q approaches P

m_{PQ} approaches $m = 1$

$$\lim_{Q \rightarrow P} m_{PQ} = m \quad \text{and} \quad \lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$$

Observe that $Q \neq P$ since in this case $m_{PQ} = 0$ and it is obvious from the graph that it is not possible.

* The velocity problem:

If $y = f(t)$ represents the position of an object at time t ($t \geq 0$) then

$$\frac{f(b) - f(a)}{b - a} = \frac{\text{change of position}}{\text{time elapsed}}$$

represents the average velocity of this object over the interval $[a, b]$

Example: The position of a particle at time t (in seconds) is given by

$$f(t) = e^t \quad (\text{in mm})$$

(a) Determine the velocity of the particle for $0 \leq t \leq 1$

(b) what is the velocity of the particle at $t=0$

Pay attention to the link between the tangent and velocity problem!!!

average velocity = slope of the secant line PQ ($Q(1, e)$)

velocity at $t=0$ or instantaneous velocity } = slope of the tangent line at P

$$\therefore (a) \text{ average velocity} = \frac{f(1) - f(0)}{1} = e - 1 \approx 1.718 \text{ mm/s}$$

(b) velocity at $t=0 \approx 1 \text{ mm/s}$ (initial velocity of the particle)